

The Epistemic Interface Problem in Multi-Agent Systems

Why Latent-Space Communication Requires a Formal Contract

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Andrei Dinga-Reassi

Independent Researcher

github.com/dravitch

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Based on: RecursiveMAS [?] · IB [?] · GIB [?] · Deep EK-NN [?] · Epistemic DL [?] · IoA [?] · Perry & Tsoukias [?] · Smets [?] · Belnap [?] · Denoëux [?]

Key changes v0.1 → v0.2. Section 3.3 corrected: the IB connection is reformulated as a neural-implementation phenomenon (not predicted by Tishby’s original IB). Section 5.3 strengthened: complete axiomatic justification of rules O1/O3/O5 (LCP, Klawonn & Smets 1992, Denoëux 2008). Section 5.5 added: theoretical unification of `belief_mass` and `belnap_state` via Perry & Tsoukias’s (1998) fuzzy Belnap space. Section 10 added: future extensions including latent credal channel, adaptive recursive fusion, and epistemic-measure stopping criterion.

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Abstract

We identify and formalise a fundamental gap in the multi-agent systems (MAS) literature, which we call the **Epistemic Interface Problem (EIP)**: the absence of a formal contract governing the transformation of an agent’s internal latent state into an auditable epistemic declaration. Current research conflates two distinct problems — *expressivity* (how to transmit rich internal states without loss) and *epistemic accountability* (how a receiving agent or orchestrator can formally reason about what it received). Recent work on latent-space communication (RecursiveMAS, Interlat, KVComm) solves the first problem but leaves the second entirely open.

We prove that the expressivity–accountability boundary is **structurally necessary**: any channel that maximises expressivity in the continuous latent space is provably non-auditable under standard certification frameworks (DO-178C, IEC 61508). We then formally characterise the EIP and propose a minimal interface contract — the **CLAIM schema** — that resolves the problem under bounded assumptions, using the Transferable Belief Model (TBM, Smets 1994), Belnap’s four-valued logic, and AGM belief revision.

We show that RecursiveMAS’s Gradient Death Theorem (Theorem 4.1, Yang et al. 2026) is a special case of a more general result derivable from Information Bottleneck theory — *restricted to neural implementations via softmax*, not to the original IB theory itself — and that this general result motivates the need for a formal epistemic interface at the latent/symbolic boundary. A new Section 5.5 establishes the theoretical unification of the `belief_mass` and `belnap_state` fields via Perry & Tsoukias’s (1998) fuzzy Belnap space, providing complete axiomatic foundations for all five CLAIM invariants.

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1. Introduction

1.1. The Confusion at the Heart of Inter-Agent Communication

The design of communication protocols between AI agents has, until recently, been treated as a single problem: how do agents share information efficiently? This framing masks a fundamental dichotomy.

Consider two agents A_1 and A_2 collaborating on a classification task. A_1 has processed sensor data and formed an internal representation $h \in \mathbb{R}^{d_h}$ encoding its assessment. It must communicate this to A_2 and to an orchestrator O that must synthesise the outputs of multiple agents into a decision. Two distinct questions arise:

Q1 — Expressivity. What proportion of A_1 ’s internal state h can be faithfully transmitted to A_2 ? What is the information loss induced by the channel?

Q2 — Epistemic accountability. What does A_2 (or O) *know* after receiving A_1 ’s message? Can O formally propagate uncertainty, detect contradictions between agents, identify the agent responsible for a conflict, and distinguish “I observed nothing” from “I did not look”?

The existing literature addresses Q1 extensively but Q2 almost entirely in isolation from Q1. No work formalises the interface between the answer to Q1 (a latent-space communication mechanism) and the answer to Q2 (a formal epistemic reasoning framework). This interface — the *Epistemic Interface* — is the subject of this paper.

1.2. Why This Is Urgent

Three convergent developments make this formalisation urgent:

Development 1 — Latent communication architectures are maturing. RecursiveMAS [?], Interlat [?], KVComm [?], and SDE [?] have demonstrated that agents communicating via latent states outperform textual MAS by 8–17% with significant efficiency gains. This architecture is becoming standard.

Development 2 — Deployment contexts demand auditability. As MAS expand into high-stakes domains — medical diagnosis, autonomous systems, military ISR (Intelligence, Surveillance, Reconnaissance), financial compliance — formal auditability, provenance tracing, and uncertainty quantification become non-negotiable. These requirements are formalised in DO-178C (avionics software), IEC 61508 (functional safety), and NATO STANAG protocols.

Development 3 — The gap is not being closed. Protocols such as A2A, MCP, and FIPA-ACL standardise message syntax but not epistemic content. Uncertainty quantification methods for LLMs operate in language space and inherit its expressive limitations. Deep evidential learning methods [? ?] produce formal belief functions but only within individual agents, not across agent boundaries in a MAS.

1.3. Contributions

This paper makes five contributions:

- Formalisation of the EIP.** We define the EIP precisely as the problem of constructing a formal mapping $\gamma : \mathbb{R}^{d_h} \rightarrow \mathbb{B}(\Theta)$ from agents’ latent states to belief structures over a shared frame of discernment Θ , under auditability, provenance, and qualified silence constraints.
- Generalisation of RecursiveMAS Theorem 4.1.** We show that the Gradient Death result is a special case of a more general theorem derivable from Information Bottleneck theory, restricted to neural implementations via softmax — not to the IB theory itself.
- Structural necessity of the Expressivity/Accountability boundary.** We prove that no single channel can simultaneously maximise expressivity (continuous latent) and satisfy auditability constraints (symbolic, discrete, certifiable).

4. **The CLAIM schema as a formal solution.** A minimal interface contract with five invariants satisfying the EIP under bounded assumptions.
5. **Theoretical unification.** We establish that the `belief_mass` and `belnap_state` fields are projections of a single mathematical object — a mass function over Perry & Tsoukias’s (1998) fuzzy Belnap space — providing complete axiomatic foundations for the CLAIM design choices.

2. Background and Related Work

2.1. Latent-Space Communication Between Agents

Textual MAS. The dominant paradigm for LLM-based MAS uses natural language as the communication medium (AutoGen, ChatDev, MetaGPT). This offers human interpretability but suffers from documented limitations: semantic compression, information loss at sampling, and inability to transmit complete reasoning trajectories.

Latent communication. A growing body of work replaces text with latent representations: CIPHER transmits probability-weighted token embeddings; AC transmits last-token hidden states; Interlat extends this to full inter-agent communication with compression; KVComm shares key-value caches; SDE transmits state-delta trajectories. RecursiveMAS [?] — the most systematic — treats the entire MAS as a recursive computation in latent space with co-optimisation via shared gradient credit attribution, demonstrating +8.3% accuracy, $1.2\times$ – $2.4\times$ speedup, and 34.6%–75.6% token reduction.

The theoretical gap. None of this work addresses what a receiving agent or orchestrator can formally conclude from the received latent state. The message is richer — but the epistemic framework for processing it is no more developed than for text.

2.2. Epistemic Frameworks for Individual Agents

Bayesian approaches. Standard uncertainty quantification in LLMs uses probability theory, typically detecting epistemic uncertainty via log-likelihood thresholding or iterative prompting. These methods remain in language space and inherit its gradient limitations.

Evidential approaches (Dempster-Shafer / TBM). DST [?] and the TBM [?] represent epistemic uncertainty as mass functions $m : 2^\Theta \rightarrow [0, 1]$, enabling explicit representation of ignorance (mass on Θ) and conflict (mass on $\emptyset$ in the unnormalised formulation). Key results: Deep EK-NN [?] applies k -nearest-neighbours in latent space to produce mass functions by distance-weighted evidential combination, explicitly separating epistemic uncertainty (non-specificity, Dubois & Prade) from aleatoric uncertainty (discord, Klir & Ramer). Epistemic Deep Learning [?] produces mass functions via random set neural networks.

Critical limitation. These frameworks produce formal belief functions *within* individual agents. They do not define how these belief functions should be communicated across agent boundaries in a MAS, nor how an orchestrator should combine belief functions from multiple agents while preserving formal epistemic properties.

2.3. Communication Protocols for MAS

FIPA-ACL (2002), A2A (2025), and MCP standardise message envelopes, performatives, and interaction patterns. They impose no agreement on semantic content or epistemic representation. The Internet of Agents architecture [?] proposes semantic layer L9 for shared ontology negotiation

— an advance toward the EIP, but it remains agnostic about belief representation, uncertainty, and conflict. No existing protocol specifies: (a) how an agent expresses its degree of belief in a proposition, formally and comparably across heterogeneous agents; (b) how a receiving orchestrator should combine belief expressions from multiple agents; (c) how to represent qualified silence; (d) how to preserve provenance so that the origin of a conflicting belief can be identified.

3. The Gradient Death Theorem — Generalisation

3.1. RecursiveMAS Theorem 4.1 — The Original Result

Theorem 3.1 (Yang et al. 2026 — Gradient Death in Textual MAS). *Under Assumption A.1 (textual SFT uses $R_{\text{text}}(h) = W_{\text{in}} \cdot \text{softmax}(W_{\text{out}} \cdot h)$ as the recursive link, with $\|W_{\text{in}}\|_2, \|W_{\text{out}}\|_2 \leq O(1)$), if tokens are confident with entropy $\varepsilon \leq \bar{\varepsilon}$ where $\bar{\varepsilon} \ll 1$, then:*

$$\left\| \frac{\partial R_{\text{text}}(h)}{\partial h} \right\|_2 \leq O(\varepsilon) \ll 1.$$

Proof sketch (verified directly in Appendix A.3 of Yang et al. 2026). (1) By chain rule: $J_{\text{text}} = W_{\text{in}} \cdot S \cdot W_{\text{out}}$, where $S = \text{diag}(p) - pp^\top$ and $p = \text{softmax}(W_{\text{out}}h)$. (2) By sub-multiplicativity: $\|J_{\text{text}}\|_2 \leq O(\|S\|_2)$. (3) Since S is the covariance matrix of a categorical distribution (PSD): $\|S\|_2 \leq \text{Tr}(S) = 1 - \|p\|_2^2$. (4) By the logarithmic inequality $\ln z \leq z - 1$: $\varepsilon \geq \sum_i p_i (-\ln p_i) \geq 1 - \|p\|_2^2$. (5) Therefore $\|J_{\text{text}}\|_2 \leq O(\varepsilon)$. $\square$

Remark 3.2 (Scope correction, v0.2). The theorem assumes a standard softmax decoder ($T = 1$, deterministic). For temperature $T \neq 1$, the bound becomes $O(\varepsilon/T)$. For stochastic decoders (top- k , top- p), the gradient is undefined in the classical sense. The theorem does not cover stochastic communication channels.

3.2. The General Theorem — Any Discrete Communication Channel

We prove that gradient death holds for any communication channel that discretises a continuous latent state.

Definition 3.3 (Discrete Communication Channel). A communication channel $C : \mathbb{R}^{d_h} \rightarrow \Delta(V)$ is a *discrete communication channel* if it maps a latent state h to a probability distribution over a finite vocabulary V with $|V| < \infty$. The effective Jacobian of the round-trip communication cycle (encode $\rightarrow$ decode) is $J_C = \partial R_C(h) / \partial h$, where $R_C(h) = \mathbb{E}_{v \sim C(h)}[\text{embed}(v)]$ is the expected re-embedding of the sampled output.

Theorem 3.4 (Generalised Gradient Death). *Let $C : \mathbb{R}^{d_h} \rightarrow \Delta(V)$ be any discrete communication channel with output distribution $p_C(h)$. Let $H(p_C(h)) \leq \varepsilon$ be the output entropy of the channel. Under bounded embedding norms, with probability $\geq 1 - \delta$:*

$$\left\| \frac{\partial R_C(h)}{\partial h} \right\|_2 \leq O(\varepsilon).$$

Proof sketch. The Jacobian decomposes as $J_C = E_{\text{embed}} \cdot S_C \cdot \nabla_h p_C$, where $S_C = \text{diag}(p_C) - p_C p_C^\top$ is the covariance matrix of the output distribution. Since S_C is PSD: $\|S_C\|_2 \leq \text{Tr}(S_C) = 1 - \|p_C\|_2^2$. By the same logarithmic inequality: $1 - \|p_C\|_2^2 \leq H(p_C) \leq \varepsilon$. Under bounded embedding norms: $\|J_C\|_2 \leq O(\varepsilon)$. $\square$

Key intuition. The entropy ε of the channel’s output distribution — not the vocabulary size, not the encoding mechanism — determines the gradient bound. The more confident an agent is ($\varepsilon \rightarrow 0$), the more the gradient vanishes, regardless of the vocabulary, the implementation of the mapping $h \rightarrow V$, or whether the vocabulary represents tokens, ontology concepts, or JSON schema values.

3.3. Connection with Information Bottleneck Theory (v0.2 — Corrected)

Correction v0.1 $\rightarrow$ v0.2. In v0.1, the connection to IB was presented as if the matrix $S = \text{diag}(p) - pp^\top$ appeared explicitly in Tishby (1999). This is incorrect. Tishby’s IB optimises directly over the simplex of probability distributions, without softmax parametrisation. The matrix S is an artefact of the *neural implementation* of IB — not of the IB theory itself. This correction strengthens Theorem 6.1: the Expressivity/Accountability boundary is architectural, not tied to an implementation choice.

In the Information Bottleneck framework [?], a communication channel corresponds to the stochastic mapping $p(T|X)$, where X is the input (latent state h), T is the transmitted representation, and Y is the downstream prediction target. The IB objective is:

$$\min_{p(T|X)} \mathcal{L}_{\text{IB}} = I(X;T) - \beta \cdot I(T;Y).$$

The matrix $S = \text{diag}(p) - pp^\top$ does not appear in the original IB formulation. It appears *only* when the IB channel $p(T|X)$ is parametrised by a neural network with a softmax layer. This is an artefact of the softmax parametrisation. RecursiveMAS resolves the problem by replacing the discrete-softmax channel with a continuous residual channel (RecursiveLink) whose Jacobian remains close to the identity — equivalent to solving IB without saturation, consistent with the GIB principle [?] of synergistic feature combination.

Corollary 3.5 (Scope of Theorem 4.1). *The Gradient Death result of Yang et al. (2026) applies to any discrete channel — including JSON-encoded ontologies, FIPA-ACL performatives, and symbolic logic assertions — when these channels are implemented via softmax on a categorical output distribution in a high-confidence regime. It does not apply to: (a) continuous latent channels (RecursiveMAS); (b) stochastic decoders (top-k, top-p); (c) algebraic operations on probability masses (TBM combination rules are not communication channels in the sense of Definition 3.1).*

Corollary 3.6 (Structural Necessity of the Boundary). *No single channel can simultaneously: (a) avoid gradient death (requires continuous latent space), and (b) satisfy formal auditability constraints (requires symbolic, discrete, certifiable representations). The boundary between the latent domain and the epistemic domain is structurally necessary — not a design choice.*

4. The Epistemic Interface Problem

4.1. Formal Definition

Definition 4.1 (Frame of Discernment). $\Theta = \{\theta_1, \dots, \theta_n\}$ is a finite set of mutually exclusive and exhaustive hypotheses, shared by all agents in the MAS and by the orchestrator.

Definition 4.2 (Belief Mass Function). A belief mass function $m : 2^\Theta \rightarrow [0,1]$ satisfies $\sum_{A \subseteq \Theta} m(A) = 1$. In the unnormalised TBM formulation [?], $m(\emptyset) \geq 0$ is the conflict mass, preserved rather than redistributed — avoiding Zadeh’s paradox.

Definition 4.3 (Belnap State). A Belnap state $\beta \in \{T, F, B, N\}$ [?] represents: T (True) = proposition believed; F (False) = proposition disavowed; B (Both) = contradictory evidence received; N (Neither) = no evidence received — the qualified-silence state.

Definition 4.4 (Epistemic Interface Problem (EIP)). Given agent A_i with internal latent state $h_t \in \mathbb{R}^{d_h}$ at time t , and an orchestrator O operating over Θ , the EIP is the problem of constructing a mapping $\gamma_i : \mathbb{R}^{d_h} \times T \times K \times P \rightarrow \mathbb{B}(\Theta)$ satisfying simultaneously: (1) **Calibration** — m is a formally calibrated belief function; (2) **Qualified silence** — the system can distinguish “no evidence” ($\beta = N$) from “negative evidence” ($\beta = F$); (3) **Provenance** — the belief function carries a verifiable identifier of its source agent; (4) **Temporal validity** — m is accompanied by a validity window; (5) **Auditability** — the belief function and its combination rules are formally certifiable.

4.2. Why the EIP Cannot Be Resolved in the Latent Domain

Proposition 4.5. *No continuous latent communication channel simultaneously satisfies constraints (2) Qualified Silence, (3) Provenance, and (5) Auditability of Definition 4.4.*

Argument. Constraint (5) requires a deterministic, symbolic, certifiable function — real-valued vectors are not directly interpretable by certification frameworks requiring discrete, enumerable, reproducible states. Constraint (2) requires a ternary distinction impossible to represent via near-zero feature values without additional symbolic scaffolding. Constraint (3) requires that `chain_id` be explicitly attached — latent vectors carry no such metadata in existing architectures. $\square$

4.3. Why the EIP Cannot Be Resolved by Current Protocols

Proposition 4.6. *No existing agent communication protocol (FIPA-ACL, A2A, MCP, SLIM) resolves the EIP. In particular: FIPA-ACL includes an *ontology* field but provides no mechanism to negotiate or validate belief mass distributions; A2A standardises task delegation but assumes agents are “honest about their uncertainty” without specifying a valid uncertainty representation; MCP shares context and allows tool calls but does not specify the epistemic content of tool outputs; none of these protocols supports $\beta = N$ (qualified silence) as distinct from $\beta = F$ (negative observation); none supports $m(\emptyset)$ as an explicit conflict indicator; the IoA L9 layer [?] advances toward semantic negotiation but remains agnostic about belief representation.*

5. The CLAIM Schema as a Formal Solution

5.1. Definition

Definition 5.1 (CLAIM — Contextual Latent Attestation of Indexed Meaning). A CLAIM is a 6-tuple:

- **proposition:** Φ — propositional content;
- **belief_mass:** $m : 2^\Theta \rightarrow [0, 1]$ — unnormalised TBM mass function, $m(\emptyset)$ = explicit conflict mass;
- **belnap_state:** $\beta \in \{T, F, B, N\}$ — four-valued epistemic state;
- **illocution:** $\kappa \in \{\text{OBSERVE, INFER, DEDUCE, ASSUME}\}$ — speech act type;
- **freshness:** $(t_{\text{obs}}, \Delta t_{\text{valid}})$ — timestamp + validity window;
- **provenance:** `chain_id` — PROV-O compatible identifier.

5.2. EIP Satisfaction

Theorem 5.2 (CLAIM Satisfies the EIP). *Under Assumption 5.1 (the mapping $\gamma_i : \mathbb{R}^{d_h} \rightarrow \text{CLAIM}$ is implemented by a calibrated, deterministic function for each agent A_i), the CLAIM schema satisfies the five constraints of Definition 4.4.*

Proof. (1) Calibration: via TBM mass function on annotated Θ data. (2) Qualified silence: $\beta = N$, explicitly distinct from B, T, F . (3) Provenance: `chain_id` implementing PROV-O lineage. (4) Temporal validity: DST discounting $m_{\text{disc}}(A) = \alpha \cdot m(A) + (1 - \alpha) \cdot m(\Theta)$. (5) Auditability: the CLAIM is a stateless, deterministic data structure whose combination rules are algebraic operations with formal proofs. $\square$

5.3. Orchestrator Rules — Axiomatic Justification (v0.2)

Rule O1 — TBM Conjunctive Combination [?].

$$m_{1 \oplus 2}(A) = \sum_{B \cap C = A} m_1(B) \cdot m_2(C), \quad A \subseteq \Theta.$$

Axiomatic justification (Klawonn & Smets 1992; Pichon & Denoëux 2008): the unnormalised conjunctive rule is the *unique* rule satisfying simultaneously (a) associativity, (b) commutativity, (c) neutral element (the vacuous belief function), (d) the Least Commitment Principle (LCP) — it never adds more belief than the available information justifies. Dempster’s normalised rule satisfies (a)–(c) but violates LCP by redistributing $m(\emptyset)$ (Zadeh’s paradox). Rule O1 is therefore axiomatically necessary, not arbitrary. The conflict mass $m(\emptyset)$ is preserved as an explicit indicator of inter-agent disagreement.

Rule O3 — Source Deduplication (Cautious Rule, Denoëux 2008). Activated when common `chain_id` prefixes detect a shared source between agents. The conjunctive cautious rule is idempotent: $m \oplus_{\text{cautious}} m = m$. This property is necessary when agents share an undeclared common source: combining the same evidence twice must not double-count it. The combination O1 (independence) + O3 (dependence) provides complete coverage of the combination problem. To our knowledge, this Smets/Denoëux pair has not been explicitly formulated as an adaptive combination policy based on detected provenance in the existing literature.

Rule O5 — Human Escalation Threshold. When $m(\emptyset) \geq \theta_{\text{conflict}}$, the orchestrator suspends automated decision-making and escalates to a human operator rather than resolving the conflict algorithmically. This is not a conservative choice — it is a consequence of the LCP. Redistributing $m(\emptyset)$ would violate LCP (adding unjustified belief). The only decision compatible with LCP in the face of persistent conflict is to expose it to the decision-maker. Escalation transmits: the value of $m(\emptyset)$, the source pair identified by PCR5, Belnap states of all involved agents, and illocution types (to distinguish conflicts between observations, inferences, or assumptions).

5.4. The Calibration Problem — The Open Question

Theorem 5.1 proves that CLAIM satisfies the EIP under Assumption 5.1 — that the mapping $\gamma_i : \mathbb{R}^{d_h} \rightarrow \text{CLAIM}$ is calibrated and deterministic. This assumption is non-trivial and represents the main open problem identified by this paper.

Deep EK-NN [?] provides the most complete answer for classification tasks: use k -nearest-neighbours in latent space to construct mass functions by distance-weighted evidential combination. This approach: (✓) produces calibrated mass functions from latent representations; (✓) explicitly separates epistemic from aleatoric uncertainty; (✓) works with pretrained models; (×) requires a labelled reference set in latent space; (×) does not extend to open-vocabulary generation tasks; (×) does not address the inter-agent boundary.

5.5. Theoretical Unification via the Fuzzy Belnap Space (v0.2 — New)

5.5.1 The Fuzzy Belnap Space (Perry & Tsoukias 1998)

Perry & Tsoukias [?] define a continuous extension of Belnap’s four-valued logic. The truth-value space becomes the set of convex combinations of the four reference values:

$$\mathbb{M} = \{(t, f, k, u) \in [0, 1]^4 \mid t + f + k + u = 1\}$$

where t = degree of truth, f = degree of falsity, k = degree of contradiction (both), u = degree of unknown (neither). The four classical Belnap values are the corners of the simplex: $T = (1, 0, 0, 0)$, $F = (0, 1, 0, 0)$, $B = (0, 0, 1, 0)$, $N = (0, 0, 0, 1)$.

5.5.2 Isomorphism with TBM Mass Functions

A mass function m over $\Theta_{\text{Belnap}} = \{T, F, B, N\}$ assigns masses to the 16 subsets of $\{T, F, B, N\}$. The atomic masses — $m(\{T\})$, $m(\{F\})$, $m(\{B\})$, $m(\{N\})$ — are exactly the components (t, f, k, u) of the Perry-Tsoukias space when restricted to masses on singletons. This is not an analogy — it is an isomorphism: every point of $\mathbb{M}$ corresponds to a belief mass function concentrated on the four Belnap atoms. The unified object is:

$$m_{\text{meta}} : 2^{\Theta \times \{T, F, B, N\}} \rightarrow [0, 1]$$

This object simultaneously encodes: (a) what the agent believes about the world (hypotheses $\theta \in \Theta$), and (b) the agent’s epistemic status regarding its own belief (truth, falsity, contradiction, ignorance).

5.5.3 Projections and CLAIM Fields

The two CLAIM fields are projections of m_{meta} :

- **belief_mass** (world level): $m_{\text{world}}(\theta) = \sum_{\beta} m_{\text{meta}}(\theta, \beta)$ — marginalisation over epistemic status.
- **belnap_state** (epistemic level): $m_{\text{epist}}(\beta) = \sum_{\theta} m_{\text{meta}}(\theta, \beta)$ — marginalisation over world hypotheses.

The discrete $\beta \in \{T, F, B, N\}$ is the limit case where m_{epist} is concentrated on a single atom. The full m_{meta} object is an architectural option for NMT 4–5.

5.5.4 Connection with O-Information

A remarkable connection with the synergy measure (GIB, [?]) emerges:

O-Information	Belnap State	TBM Interpretation
+1 (max synergy)	T (true)	Pure complementarity — information combines perfectly
−1 (max redundancy)	F (false)	Total overlap — agents duplicate
0 with mixed components	B (contradiction)	Tension — synergy and redundancy coexist
0 with null components	N (ignorance)	Independence — no shared information

5.5.5 Adaptive Stopping Criterion for RecursiveMAS

The fuzzy Belnap representation immediately suggests a principled stopping criterion for RecursiveMAS recursive rounds:

Stopping criterion: Terminate recursion when u (unknown) and k (contradiction) fall below their respective thresholds θ_u and θ_k — indicating that belief has stabilised and additional rounds will no longer reduce epistemic uncertainty.

Escalation criterion: If k (contradiction) remains persistently above θ_k despite $r \geq r_{\max}$ rounds, trigger Rule O5 (human escalation) — the system has identified an irresolvable inter-agent conflict.

This criterion is formally justified by the axiomatic structure of the CLAIM: stopping when $u \downarrow$ and $k \downarrow$ is a direct application of the LCP (believe only what the information justifies; stop when no new information arrives). To our knowledge, no existing MAS paper proposes a formally justified stopping criterion for recursive collaboration based on epistemic state.

6. The Expressivity / Accountability Boundary

Theorem 6.1 (Structural Necessity of the Expressivity/Accountability Boundary). *No communication channel $C : \mathbb{R}^{d_h} \rightarrow O$ simultaneously satisfies: (a) **Gradient preservation:** $\exists c > 0, \forall h \in M, \|\partial R_C / \partial h\|_2 \geq c$ for all entropy values ε ; and (b) **Formal auditability:** O is a discrete, enumerable, certifiable output space.*

Proof. By Theorem 3.1, if O is discrete (condition b), then $\|\partial R_C / \partial h\|_2 \leq O(\varepsilon)$, which vanishes as $\varepsilon \rightarrow 0$ (violates condition a). If O is continuous (satisfies condition a via RecursiveLink), then $O = \mathbb{R}^{d'}$ is not a discrete, enumerable space (violates condition b). $\square$

Continuous Latent Domain	Symbolic Epistemic Domain
RecursiveMAS, Interlat, KV-Comm $h_t \in \mathbb{R}^{d_h}$	CLAIM + Orchestrator $m : 2^\Theta \rightarrow [0, 1], \beta \in \{T, F, B, N\}$
Gradient preservation	Certifiable, auditable
Expressivity maximisation	Accountability maximisation
Opaque to certification	DO-178C / IEC 61508 compatible
$\longleftrightarrow \gamma_i \longrightarrow$ (open calibration problem — Section 5.4)	

7. Related Work — Positioning

Work	Expressivity	EIP	Across Agent Boundaries
RecursiveMAS [?]	✓ Complete	×	N/A (latent domain)
Interlat [?]	✓ Partial	×	N/A
Deep EK-NN [?]	N/A	✓ Single agent	×
Epistemic DL [?]	N/A	✓ Single agent	×
FIPA-ACL / A2A / MCP	×	×	Syntactic only
IoA L9 [?]	N/A	~ Partial (ontology)	✓ Non-epistemic
This work — CLAIM	N/A (by Thm. 6.1)	✓ Formal	✓ Defined

8. Conclusion

We have identified, formalised, and partially resolved the Epistemic Interface Problem in multi-agent systems. The central contributions of v0.2 are:

1. The EIP is a problem distinct from expressivity. Improving the communication channel (RecursiveMAS) does not resolve the EIP. Improving epistemic frameworks within individual agents (Deep EK-NN) does not resolve the EIP. The interface between the two domains is the gap.
2. Gradient death is more general than RecursiveMAS claims — it applies to any discrete channel parametrised by softmax — but is not predicted by the IB theory itself. This correction strengthens the paper’s argument by making it implementation-agnostic at the architectural level.
3. The Expressivity/Accountability boundary is necessary (Theorem 6.1). Hybrid neuro-symbolic architectures must make this boundary explicit.
4. CLAIM is a minimal formal solution satisfying the EIP under bounded assumptions. The calibration problem (γ_i) is the open research frontier.
5. The `belief_mass` and `belnap_state` fields are theoretically unified: they are projections of a single mass function over Perry & Tsoukias’s (1998) fuzzy Belnap space. This provides complete axiomatic foundations for all CLAIM invariants and for orchestrator rules O1/O3/O5.

9. Open Problems

- Formal convergence proof for the proposed γ_i calibration (Section 5.4)
- Extension to open-world hypothesis spaces ($|\Theta|$ not predefined)
- Empirical validation on multimodal ISR datasets
- Integration with formal verification tools for orchestrator combination rules
- Strategic behaviour analysis: what happens if agents have incentives to misreport their `belief_mass`?
- Formal study of the Smets/Denoœux duality (O1/O3) as an adaptive combination policy

10. Future Extensions

Three concrete research directions follow from the theoretical unification of Section 5.5.

10.1. Latent Channel with Credal Output

Replace RecursiveMAS’s final text output with a CLAIM: the last RecursiveLink produces not a decoded text but a quadruplet (t, f, k, u) in the fuzzy Belnap space — a continuous credal output. Residual epistemic uncertainty is measured by u (unknown) and k (contradiction). This architecture is the direct implementation of Corollary 3.2: the continuous latent domain (RecursiveMAS) feeds the symbolic epistemic domain (CLAIM) via the calibrated mapping γ_i .

10.2. Recursive Belief Fusion

After each RecursiveMAS round, combine all agents’ CLAIM mass functions by an adaptive rule: Rule O1 (Smets, unnormalised conjunctive) for independent agents (distinct `chain_id` prefixes); Rule O3 (Denoœux cautious) for source-correlated agents (common `chain_id` prefixes). This adaptive fusion transforms RecursiveMAS from a purely latent architecture into an explicit epistemic reasoning system. The accumulated conflict mass $m(\emptyset)$ across rounds is the signal for both additional rounds (if below θ_{conflict}) and human escalation (if above θ_{conflict}).

10.3. Adaptive Stopping via Epistemic Uncertainty Measure

Using the quadruplet (t, f, k, u) as the agent state, define a formally justified stopping criterion:

- Continue recursion when $u + k > \theta_{\text{uncertainty}}$
- Stop when $u + k \leq \theta_{\text{uncertainty}}$ for all agents (belief stabilised)
- Escalate when $k > \theta_{\text{conflict}}$ for $r \geq r_{\text{max}}$ consecutive rounds

This is the first formally justified adaptive stopping rule for recursive multi-agent collaboration, grounded in TBM axiomatics and the LCP. It replaces RecursiveMAS's fixed-round approach ($r = 1, 2, 3$) with a principled termination condition.

Integration path: NMT 2–3: operational CLAIM schema with discrete Belnap (current specification). NMT 4–5: full $m_{\text{meta}} : 2^{\Theta \times \{T, F, B, N\}} \rightarrow [0, 1]$ with continuous fuzzy Belnap output, adaptive stopping and recursive belief fusion.

References